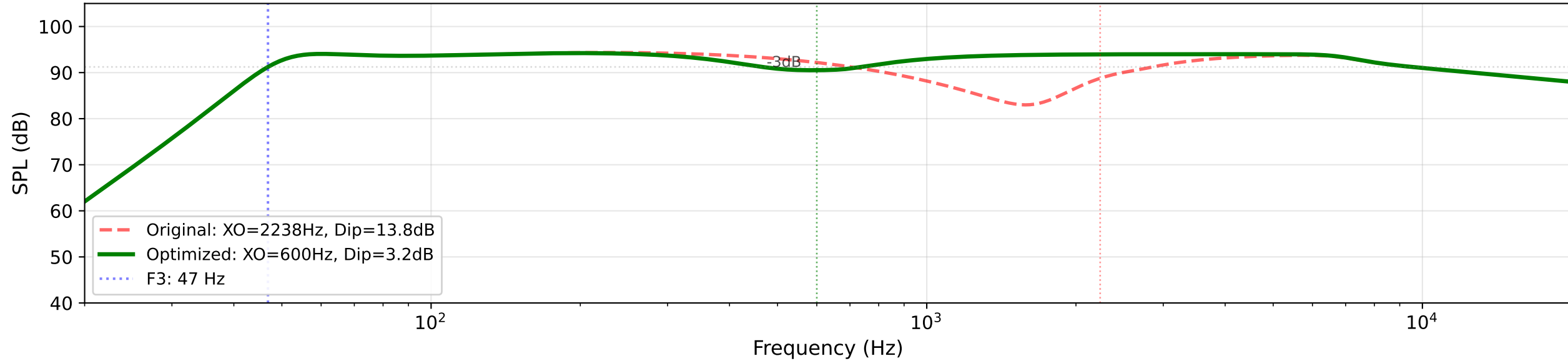
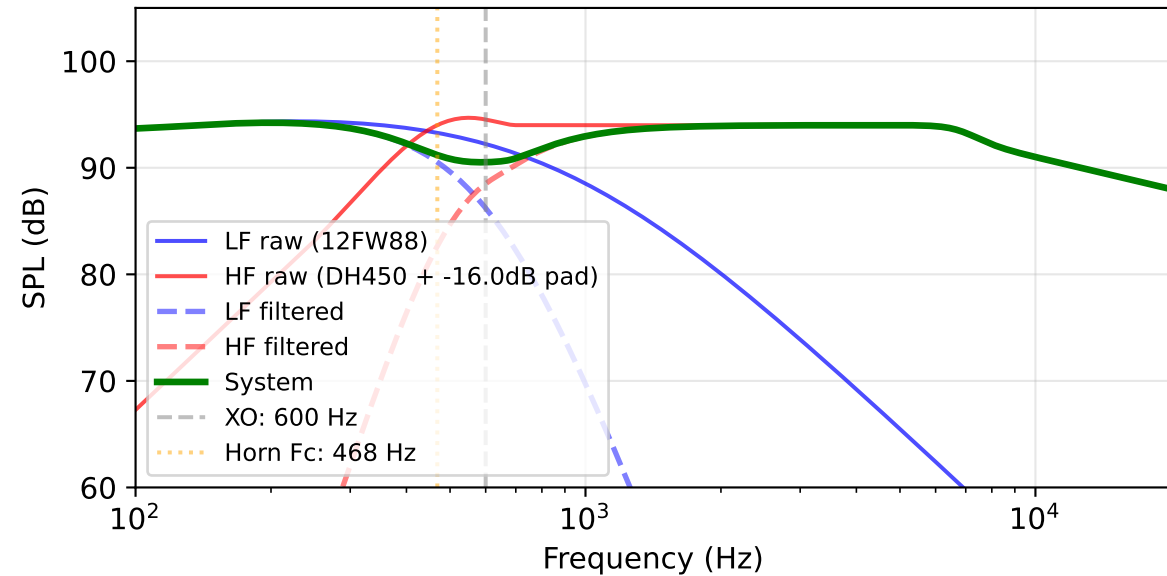


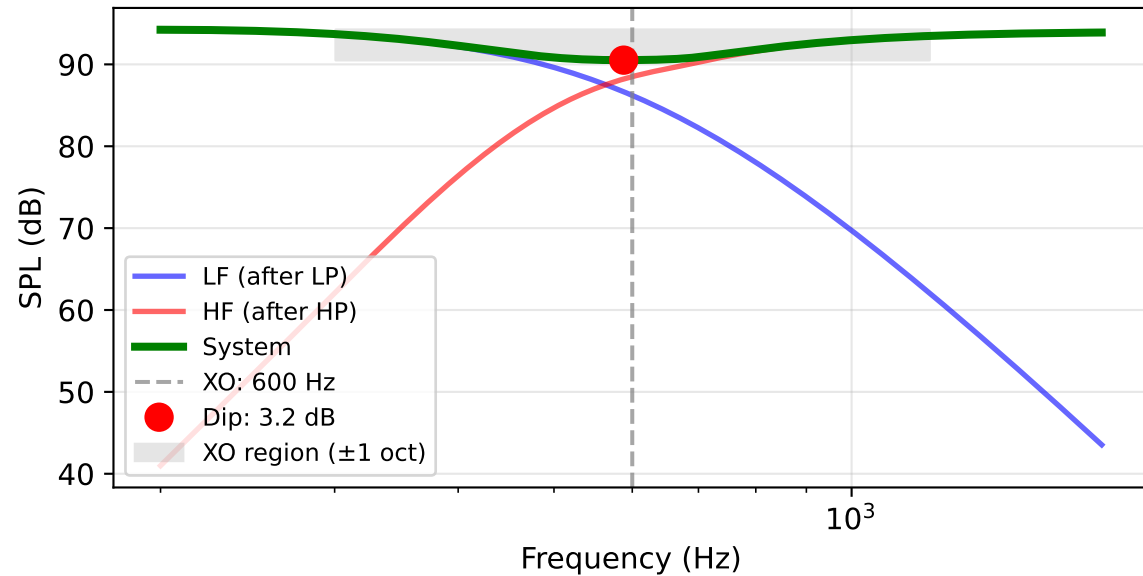
System Response Comparison: Original vs Optimized Horn



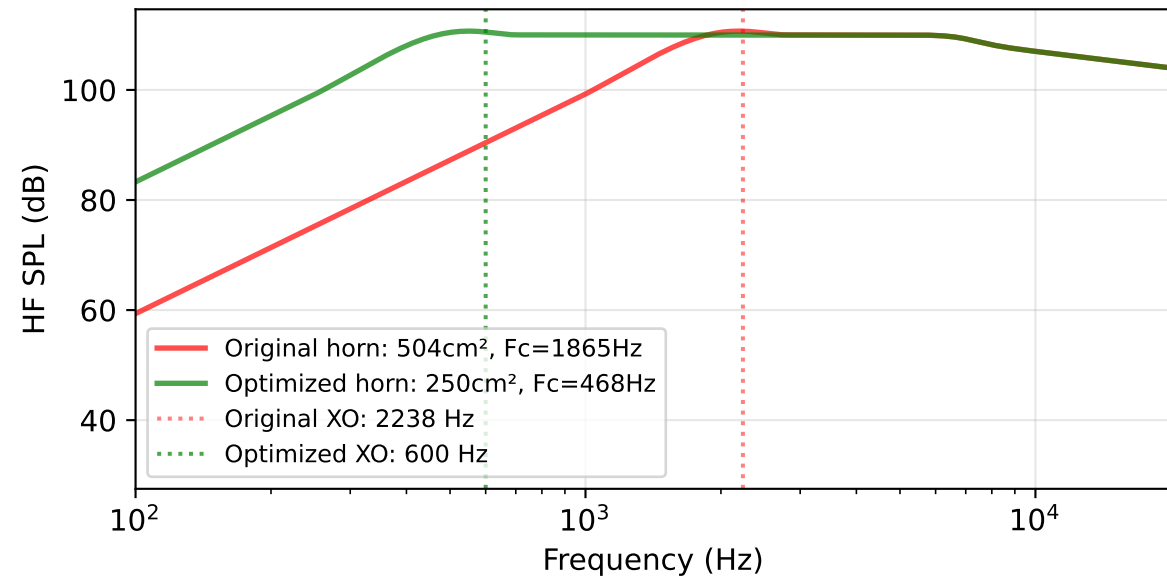
Optimized System: Individual Driver Responses



Crossover Region Detail (200-1800 Hz)



Horn Response Comparison (raw, before padding)



OPTIMIZED SYSTEM METRICS

Horn Design:

- Mouth area: 250 cm² (-50.4% vs original)
- Horn Fc: 468 Hz (-74.9% vs original)
- Length: 25 cm (250 mm)

Crossover:

- Frequency: 600 Hz (1.28×Fc)
- Type: LR4 (4th-order Linkwitz-Riley)
- HF padding: -16.0 dB

Performance:

- Crossover dip: 3.18 dB ✓ (was 13.75 dB)
- Flatness: 3.72 dB ✓ (was 11.36 dB)
- System F3: 47 Hz (maintained)
- Rating: △ Acceptable (borderline Good)

Improvement vs Original:

- Dip reduced: 13.75 → 3.2 dB (+10.6 dB)
- Flatness: 11.36 → 3.7 dB (+7.6 dB)
- XO frequency: 2238 → 600 Hz (-1638 Hz)